

# 6DOF Arm — Wiring Bench Sheet

ESP32 + PCA9685 + 6× MG996R · print & keep at the bench · full guide: [hardware/wiring.md](#)

## SAFETY RULES — read before plugging anything in

- 1. NEVER put 12 V on the PCA9685 V+ terminal or any servo.** MG996Rs are rated 4.8–7.2 V. The 12 V supply feeds **ONLY** the converter input.
- 2. USB power = single-servo bench tests only.** Full-arm motion requires the 6 V/10 A rail.
- 3. Verify converter output  $\approx 6.0$  V with a meter BEFORE first connection.** Re-check polarity at V+ — reverse polarity kills the PCA9685 instantly.
- 4. Converter → PCA9685 wires short and thick** (14–16 AWG). Stalls can pull 15 A+.
- 5. Power-up order:** ESP32 USB first (servos stay idle), then the 12 V supply. Reverse to power down.
- 6. Pinch points:** on link loss the arm **HOLDS** (won't drop, won't yield). Kill 12 V before handling.

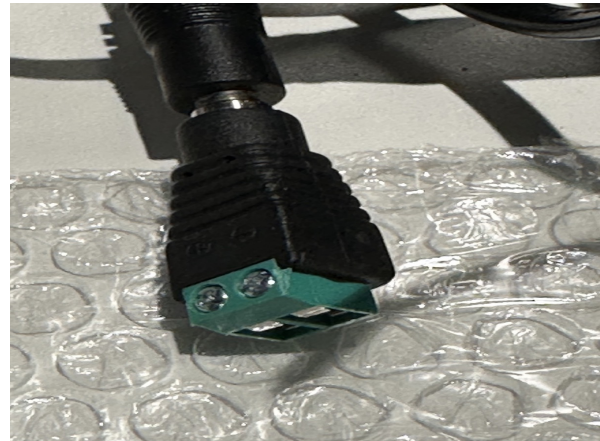
## Servo power path

12 V/10 A supply → buck converter (12–40 V in, 6 V/10 A out) → PCA9685 V+ terminal

## Converter wire colors — straight from the unit's label

| Wire                     | Role           | Connects to                       |
|--------------------------|----------------|-----------------------------------|
| Red                      | Input +        | 12 V supply positive              |
| Black (paired w/ red)    | Input –        | 12 V supply negative              |
| Yellow                   | Output + (6 V) | PCA9685 <b>V+</b> screw terminal  |
| Black (paired w/ yellow) | Output –       | PCA9685 <b>GND</b> screw terminal |

The two blacks are per-bundle negatives — use each with its own side. Red side drinks 12 V; yellow side feeds the servos. The PCA9685's onboard 1000  $\mu$ F cap covers the rail; no extra cap needed.



Left: the converter's own label (IN red/black, OUT yellow/black). Right: barrel-jack → screw-terminal adapter for the 12 V input — mind the +/– marks.

## Logic side: ESP32 ↔ PCA9685 (I2C)

PCA9685 header order, starting from the end AWAY from the silver capacitor: **GND, OE, SCL, SDA, VCC, V+** (V+ is always the pin beside the cap). Ribbon colors below are this build's harness as photographed. I2C address 0x40, jumpers open; board pull-ups are to VCC — which is why VCC must be 3.3 V.

| Ribbon color          | PCA9685 pin | ESP32 pin                      | Notes                                                                  |
|-----------------------|-------------|--------------------------------|------------------------------------------------------------------------|
| brown/black (end pin) | GND         | GND                            | <b>Required</b> — the common-ground link; I2C is dead/flaky without it |
| red                   | OE          | <b>leave unplugged</b>         | Already pulled low (enabled); future kill-switch option                |
| orange                | SCL         | D22 (GPIO22)                   | Arduino Wire default clock                                             |
| yellow                | SDA         | D21 (GPIO21)                   | Arduino Wire default data                                              |
| green                 | VCC         | 3V3                            | <b>3.3 V only</b> — never the 5 V/VIN pin                              |
| blue (beside cap)     | V+          | <b>REMOVE — never to ESP32</b> | <b>6 V servo rail!</b> Same net as the terminal block; kills the ESP32 |

**ESP32 pin locations** (30-pin devkit): USB-C toward you, antenna away — right-hand row from the USB end reads 3V3, GND, D15, D2, D4, RX2, TX2, D5, D18, D19, **D21**, RX0, TX0, **D22**, D23. Green + brown at the near corner; yellow on D21 (11th); orange on D22 (14th).

## Servo plugs and channels

Column pin order from board edge inward: **GND (bottom), V+ (middle), PWM (top)**. MG996R leads: **brown = GND, red = V+, orange = signal** — brown faces the board edge.

| Channel | Joint  | Position on arm     |
|---------|--------|---------------------|
| 0       | joint1 | Base rotation (yaw) |
| 1       | joint2 | Shoulder pitch      |
| 2       | joint3 | Elbow pitch         |
| 3       | joint4 | Wrist pitch         |
| 4       | joint5 | Wrist roll          |
| 5       | joint6 | Gripper             |
| 6–15    | —      | Spare               |

Wrong channel? Move the plug, don't edit firmware — keeps this table true.

## Common ground — required, not optional

PSU(–) → converter IN(–) → converter OUT(–) → PCA9685 GND → ESP32 GND (via the I2C ground jumper). Beep-test all four points.

## Bench bring-up checklist

- [ ] All grounds common (continuity-beep PSU–, converter–, PCA9685 GND, ESP32 GND)
- [ ] Converter output ≈ 6.0 V, verified with meter, no load
- [ ] I2C jumpers by color: green→3V3 · yellow→D21 · orange→D22 · brown→GND
- [ ] Red (OE) jumper parked; blue (V+) jumper REMOVED from ribbon
- [ ] ESP32 on USB: serial prints “PCA9685 initialized, all channels off”
- [ ] One servo on channel 0 — single-servo sweep (USB/bench power OK)
- [ ] Remaining servos on channels 1–5 per table
- [ ] Full-arm motion ONLY on converter power